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SECURITY AUDIT OF

BULLISH SOLANA SECURITY TOKEN CONTRACT V1



Public Report

Version: 1.1

July 1, 2026

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ABBREVIATIONS

Name	Description
Smart contract	A computer protocol intended to digitally facilitate, verify or enforce the negotiation or performance of a contract.
Solana	A decentralized blockchain built to enable scalable, user-friendly applications for the world.
SOL	A cryptocurrency whose blockchain is generated by the Solana platform.
Program	An application interacts with a Solana cluster by sending it transactions with one or more instructions. The Solana runtime passes those instructions to the program.
Instruction	The smallest contiguous unit of execution logic in a program.

EXECUTIVE SUMMARY

This Security Audit Report was prepared by Verichains on July 1, 2026. We would like to thank Bullish for trusting Verichains to audit the smart contracts. Delivering high-quality audits is always our top priority.

This audit focused on identifying security flaws in the code and the design of the Solana Security Token Contract. The scope of the audit is limited to the source code files provided to Verichains. Verichains completed the assessment using manual review, static analysis, and dynamic analysis techniques.

During the audit process, the audit team identified one minor issue. The Bullish team has addressed it.

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1. Management Summary

1.1. About Bullish Solana Security Token Contract v1

The Bullish Tokenization Platform is an institutional-grade token infrastructure featuring advanced compliance and security capabilities. This report covers the security token contracts on Solana. The BLSH token is the first token to use the infrastructure.

It is designed to deliver provably secure token functionality through on-chain smart contracts and programs, ensure compliance with applicable regulatory requirements, and uphold strong security standards for off-chain operations and key management.

Verichains conducted a review of the platform's Solana token implementations to evaluate whether the implementations align with its stated security and compliance objectives.

Verichains also validated the BLSH on-chain deployment to confirm its configuration and security settings are correct. See **BLSH Onchain Overview**.

1.2. About Verichains

Since 2017, Verichains has been a pioneering and leading blockchain security firm in APAC, with extensive expertise in security, cryptography, and core blockchain technology. Over 200 clients trust us to protect assets valued at more than \$50 billion, including several high-profile clients such as BNB Chain, Bybit, Aptos, Sui, Polygon, Klaytn, Wemix, Line Corp, Ronin Network, and Kyber Network.

With in-depth research and development of blockchain technology, Verichains offers a range of security services, including blockchain protocol and smart contract security audit, mobile application protection, key management solutions, on-chain risk monitoring, and red team/penetration testing services.

Verichains is a certified CREST penetration testing member, delivering professional testing services at the highest global standards.



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1.3. Audit Methodology

Our security audit process for Solana smart contracts includes two steps:

- Smart contract codes are scanned/tested for commonly known and more specific vulnerabilities using our in-house smart contract security analysis tool.
- Manual audit of the codes for security issues. The contracts are manually analyzed to look for any potential problems.

Following is the list of commonly known vulnerabilities that were considered during the audit of the Solana smart contract:

- Arithmetic Overflow and Underflow
- Signer checks
- Ownership checks
- Rent exemption checks
- Account confusions
- Bump seed canonicalization
- Closing account
- Signed invocation of unverified programs
- Numerical precision errors
- Logic Flaws

For vulnerabilities, we categorize the findings into categories as listed in the table below, depending on their severity level:

SEVERITY LEVEL	DESCRIPTION
CRITICAL	A vulnerability that can disrupt the contract functioning; creates a critical risk to the contract; required to be fixed immediately.
HIGH	A vulnerability that could affect the desired outcome of executing the contract with high impact; needs to be fixed with high priority.
MEDIUM	A vulnerability that could affect the desired outcome of executing the contract with medium impact in a specific scenario; needs to be fixed.
LOW	An issue that does not have a significant impact and can be considered less important.

Table 1: Severity levels

1.4. Disclaimer

Bullish acknowledges that the security services provided by Verichains are conducted to the best of its professional abilities but cannot guarantee 100% coverage of all security vulnerabilities. Bullish understands and accepts that despite rigorous auditing, certain vulnerabilities may remain undetected.

Therefore, Bullish agrees that Verichains shall not be held responsible or liable, and shall not be charged for any hacking incidents that occur due to security vulnerabilities not identified during the audit process.

1.5. Acceptance Minute

This final report served by Verichains to Bullish will be considered an Acceptance Minute. Within 7 days, if no further responses or reports are received from Bullish, the final report will be considered fully accepted by Bullish without the signature.

2. Audit Overview

2.1. Audit Scope

This audit focused on identifying security flaws in the code and the design of the Bullish Solana Security Token Contract v1. It was conducted on commit [4cd1676562ccea1f677a27d8a22dfa046f892ad](#) from the Git repository [bullish-stampede](#).

The latest versions of the following files were made available in the course of the review:

- Solana:

SHA256 Sum	File
524e41fad6505e28793b92e75e5428be593ecb29e807d4764dd4542f2395a302	scripts/create-mint.ts
1ac7c83342a2966dbce78dadffee812ebe4df92b7b7a50fe7a7b546b13b2cdc5	scripts/transfer-authorities.ts

Table 2: Native Solana Token-2022 Mint

2.2. Overview

2.2.1. Solana program:

This repository creates a native Solana Token-2022 mint with extensions and splits authorities between an operation key and an admin key.

The process deploys a native Token-2022 mint using [create-mint.ts](#), then transfers authorities using [transfer-authorities.ts](#).

- Mint creation:

[create-mint.ts](#) creates the mint account, initializes Token-2022 extensions, initializes the base mint, and then initializes Token Metadata on the mint.

The script enables these extensions:

- **PermanentDelegate**: allows a configured delegate to transfer or burn tokens from token accounts for this mint.
- **TransferHook**: reserves a transfer hook authority and hook program slot for future program integration. Initializes the authority but leaves the hook program unset.
- **MetadataPointer**: points token metadata to the mint account itself.
- **PausableConfig**: enables mint-level pause/resume capability through Token-2022.
- **ScaledUiAmountConfig**: enables a configurable UI display multiplier. Initializes this with multiplier **1**.
- **ConfidentialTransferMint**: enables confidential-transfer mint configuration. Initializes it with **autoApproveNewAccounts = false** and no auditor ElGamal public key.

- **DefaultAccountState**: enabled only for the whitelist variant. New token accounts start frozen by default, so accounts must be thawed before they can transfer.

- **TokenMetadata**: initialized after the mint and extensions are created. Stores name, symbol, URI, update authority, and additional metadata.

- **Authority split:**

After mint creation, [*transfer-authorities.ts*](#) transfers authorities away from the deployer.

The **operation key** receives daily operational authorities:

- **Mint authority**: can mint tokens.
- **Freeze authority**: can freeze and thaw token accounts.

The **admin key** receives governance and emergency authorities:

- **Permanent delegate**: can transfer or burn tokens from token accounts for this mint.
- **Transfer hook authority**: can set, update, or clear the transfer hook program.
- **Metadata pointer authority**: can update the account used as the metadata pointer.
- **Token metadata update authority**: can update name, symbol, URI, and additional metadata.
- **Pause authority**: can pause and resume token operations.
- **Scaled UI amount authority**: can update the UI amount multiplier.
- **Confidential transfer mint authority**: can update confidential-transfer mint settings.

At the end of the process:

- the token is a native Token-2022 mint;
- the required extensions are enabled;
- the deployer no longer owns token authorities;
- the operation key controls day-to-day mint/freeze operations;
- the admin key controls governance, emergency, and higher-risk extension authorities.

3. Solana Findings

#	Title	Severity	Status
1	Transfer Hook programId Should Use null Instead of Sentinel Address	INFO	Fixed

3.1. INFO - Transfer Hook programId Should Use null Instead of Sentinel Address - FIXED

Position

- `scripts/create-mint.ts:44` (constant), `scripts/create-mint.ts:194-198` (`getInitializeTransferHookInstruction` call)

Description

The script introduces an `UNASSIGNED_HOOK_PROGRAM` constant to represent "no transfer hook program":

```
const UNASSIGNED_HOOK_PROGRAM = address("11111111111111111111111111111111");

getInitializeTransferHookInstruction({
  mint: mintSigner.address,
  authority: deployer.address,
  programId: UNASSIGNED_HOOK_PROGRAM,
}),
```

The v2 SDK (`solana-program/token-2022`) already supports nullable `programId` (`OptionOrNullable<Address>`), so `null` should be used instead of introducing another constant.

Recommendation

```
getInitializeTransferHookInstruction({
  mint: mintSigner.address,
  authority: deployer.address,
  programId: null,
}),
```

Update

- **Apr 23, 2026:** The issue has been fixed by the development team.

4. BLSH Onchain Overview

The audited version of the Bullish Solana Security Token Contract v1 program, as of April 28, 2026, contains the following information:

PROPERTY	VALUE
Token Name	Bullish Ordinary Shares
Token Symbol	BLSH
Token Mint Address	6d5zakCaxjjRALNRyudC6AriVxeBGT3XUAci7ybWQY8U
Data URI	https://assets.bullish.com/metadata/blsh.json
Current Supply	10 (x10 ⁶)
Mint authority	4Be7JXd6JaiXd6KkSMBEya5yt19gqld11NEWTZC6Bsgk
Freeze authority	4Be7JXd6JaiXd6KkSMBEya5yt19gqld11NEWTZC6Bsgk
Created At	April 22, 2026 (UTC)
Market	Not Listed Yet
Token-2022 Program	Yes
Creator	SrwCgtpWkR97JwvuNvkfqunW83pmzCaU8nzVNtCDFaS
Create Transaction	https://solscan.io/tx/4FF4KzxqunD9QNbPQ987JtzMvM9LhbJBRkInJZg43Q9vx8Vbh8NXaqUNdnMA2323XeJpfjCU4Hh9C3m43MxoHao
Extensions	True
Holder	100% by 5HUQfMWdVkvwupcnLHmwAzRCxDhXBxHKhcK5MB7HesqK
Update Authority	DPT54eBQJf7ghEQTzX7vjAb8WaDRMJFQchX6MKm3dFcD

Table 3: Token program information

Extensions Enabled

Extension	Configuration
PermanentDelegate	Authority: DPT54eBQJf7ghEQTzX7vjAb8WaDRMJFQchX6MKm3dFcD
TransferHook	Authority: DPT54eBQJf7ghEQTzX7vjAb8WaDRMJFQchX6MKm3dFcD program_id: NONE
DefaultAccountState	state: whitelist (Frozen by default)

MetadataPointer	Authority: <code>DPT54eBQJf7ghEQtzX7vjAb8WaDRMJFQchX6MKm3dFcD</code> metadata_address: <code>6d5zakCaxjjRALNryudC6ArivxeBGT3XUAci7ybWQY8U</code>
PausableConfig	Authority: <code>DPT54eBQJf7ghEQtzX7vjAb8WaDRMJFQchX6MKm3dFcD</code> paused: false
ScaledUiAmountConfig	Authority: <code>DPT54eBQJf7ghEQtzX7vjAb8WaDRMJFQchX6MKm3dFcD</code> multiplier: 1 new_multiplier_effective_timestamp: 0 new_multiplier: 1
ConfidentialTransferMint	Authority: <code>DPT54eBQJf7ghEQtzX7vjAb8WaDRMJFQchX6MKm3dFcD</code> auto_approve_new_accounts: false auditor_elgamal_pubkey: NONE
TokenMetadata	Update Authority: <code>DPT54eBQJf7ghEQtzX7vjAb8WaDRMJFQchX6MKm3dFcD</code> mint: <code>6d5zakCaxjjRALNryudC6ArivxeBGT3XUAci7ybWQY8U</code> name: Bullish Ordinary Shares symbol: BLSH uri: https://assets.bullish.com/metadata/blsh.json additional_metadata: {}

Table 4: Extension enabled information

The system utilizes two primary keys: the Operation Key and the Admin Key. The Operation Key (`4Be7JXd6JaiXd6KkSMBEya5yt19gq1d11NEWTZC6Bsgk`) holds the Mint authority, allowing it to mint new tokens, and the Freeze authority, which can freeze and thaw token accounts. It also has the ability to transfer or revoke these authorities. The Admin Key (`DPT54eBQJf7ghEQtzX7vjAb8WaDRMJFQchX6MKm3dFcD`) acts as a permanent delegate, capable of transferring or burning tokens from any token account for this mint without the account owner's approval. Additionally, the Admin Key holds the authority to manage various Token-2022 extensions, including the Transfer Hook, Metadata Pointer, Token Metadata Update, Pause, Scaled UI Amount, and Confidential Transfer Mint authorities. It can also transfer or revoke these extension authorities.

The token has enabled several Token-2022 extensions, most of which are under the authority of the Admin Key. These include the PermanentDelegate, TransferHook (currently with no program set), MetadataPointer (pointing to the token mint address), PausableConfig (currently not paused), and ScaledUiAmountConfig (with a current multiplier of 1). The DefaultAccountState extension is also enabled and configured to "whitelist", meaning that token accounts are frozen by default.

Furthermore, the ConfidentialTransferMint extension is configured with `auto_approve_new_accounts` set to `false` and no auditor ElGamal public key. The TokenMetadata extension stores the token's name ("Bullish Ordinary Shares"), symbol ("BLSH"), and URI, with no additional metadata specified.

5. Version History

Version	Date	Status/Change	Created by
1.0	May 4, 2026	Private Report	Verichains
1.1	July 1, 2026	Public Report	Verichains

Table 5: Report versions history